

Diversifying Concentrated Revenue:

A Cohort and RFM Analysis of E-Commerce Revenue

This document is meant to explain what I did and why I did when I was cleaning, exploring, analyzing and visualizing the e-commerce dataset.

Cleaning

Cleaning was done via Excel, and since the focus was on revenue, I decided to delete any rows that were about admin expenses or returns. This also included noticing that many rows were missing CustomerIDs which could cause issues for things like Cohort Analyses, but are still valuable for a revenue analysis. The table below is the issue log from the Excel file where I cleaned the dataset. The dataset was then converted into a csv file so it could be converted into a SQLite database in the next section.

Column	Issue	Solution	Row Count	Comment
Quantity	Negative/Zero Values	Delete Rows	12327 (2009-2010) 10625 (2010-2011)	These rows were deleted so I could focus on revenue rather than lost inventory or admin expenses.
Price	Negative/Zero Values	Delete Rows	3691 (2009-2010) 2518 (2010-2011)	Same as Above
Invoice	A few Admin Cost and Returns rows were left	Delete Rows	~2	Same as Above
CustomerID	Missing Values	Left as is	103912 (2009-2010) 132219 (2010-2011)	Aside from the missing ID, these rows are perfectly fine and are left to be create the best revenue analysis.

Exploration

After creating the SQLite database, the first order of business was to look at the distribution of revenue, invoices, and average order value throughout the two year period. This was the most basic thing to look at, and it revealed the first and most important insight: revenue is in fact concentrated during fall and

December. The company does the most business during these times and their metrics are these best during this time, which means that the off-peak season during spring and summer could use marketing campaigns and other resources to be able to improve overall metrics.

RFM Analysis

RFM (Recency, Frequency, Monetary Value) is a way to rank customers to figure out who are the most valuable for the business, and then segment customers by these ranking. Recency refers to the most recent purchase, frequency to the total number of orders, and monetary value to the total value of the orders. Each of these can be multiplied by weights to give that aspect more or less weight to the customer's value to the business, but I decided to give equal weight to each of the values. This was because, recency, frequency, and monetary value are all equally valuable to an e-commerce business. Other businesses, such as one that sells expensive machinery, may not value frequency or recency as much as monetary value. There it would make sense to give monetary value a weight of 0.5, for example, and the other two a weight of 0.25 each.

The analysis revealed that the top 14% of customers, labeled Top Customers in the customer segmentation, contributed around half of the revenue. This coupled with the previous insight that the revenue was concentrated into four months, means that this company's revenue is heavily concentrated. The next top segment, High Value Customers, make up 15% of the customer base and provide a similar percentage of the revenue. The remaining segments make up 70% of the customer base and make up the remaining 35% of the revenue.

Cohort Analysis

Cohort Analyses look at how cohorts, grouped by the month and year of their first purchase, behave over time, typically seeing if they return. The first cohort is especially loyal with 30-40% of those customers returning each month, which are good numbers though I have my doubts if the cohort is accurate. I think this may be customers who first purchased in December 2009, but also loyal customers prior to this because of how anomalous the cohort is. There was also a strange down tick in activity that I tried to investigate, and I figure the issue was an increase in missing CustomerIDs skewing the data. There were about 33% more missing values during that period than the year prior.

Conclusion

My general conclusion is that revenue is concentrated in a small group of customers, and a quarter of the year which is an issue. The suggestion is to look at the off-peak period, and invest more resources into marketing then, and to launch a VIP program to keep those top customers loyal and perhaps to get customers to spend in those off-peak periods. Both of these are intended to ensure that a bad holiday season, or a top customer leaving substantially harms the revenue of the company.